

PROBLEM 01

TASK 01 : Load the provided "car_price_dataset.csv" file into a DataFrame named "df". Display the first 10 rows of the DataFrame to understand its structure.

❖ Loading the file into the DataFrame

```
import pandas as pd

df = pd.read_csv('C:/Users/Shanya/Downloads/car_price_dataset.csv')

df
```

Output :

	Brand	Model	Year	Engine_Size	Fuel_Type	Transmission	Mileage	Doors	Owner_Count	Price
0	Kia	Rio	2020	4.2	Diesel	Manual	289944	3	5	8501
1	Chevrolet	Malibu	2012	2.0	Hybrid	Automatic	5356	2	3	12092
2	Mercedes	GLA	2020	4.2	Diesel	Automatic	231440	4	2	11171
3	Audi	Q5	2023	2.0	Electric	Manual	160971	2	1	11780
4	Volkswagen	Golf	2003	2.6	Hybrid	Semi-Automatic	286618	3	3	2867
...	...	...	...	...	...	...	...	...	...	...
9995	Kia	Optima	2004	3.7	Diesel	Semi-Automatic	5794	2	4	8884
9996	Chevrolet	Impala	2002	1.4	Electric	Automatic	168000	2	1	6240
9997	BMW	3 Series	2010	3.0	Petrol	Automatic	86664	5	1	9866
9998	Ford	Explorer	2002	1.4	Hybrid	Automatic	225772	4	1	4084
9999	Volkswagen	Tiguan	2001	2.1	Diesel	Manual	157882	3	3	3342

10000 rows × 10 columns

❖ Displaying the first 10 rows of the DataFrame

```
df.head(10)
```

0	Kia	Rio	2020	4.2	Diesel	Manual	289944	3	5	8501
1	Chevrolet	Malibu	2012	2.0	Hybrid	Automatic	5356	2	3	12092
2	Mercedes	GLA	2020	4.2	Diesel	Automatic	231440	4	2	11171
3	Audi	Q5	2023	2.0	Electric	Manual	160971	2	1	11780
4	Volkswagen	Golf	2003	2.6	Hybrid	Semi-Automatic	286618	3	3	2867
5	Toyota	Camry	2007	2.7	Petrol	Automatic	157889	4	4	7242
6	Honda	Civic	2010	3.4	Electric	Automatic	139584	3	1	11208
7	Kia	Sportage	2001	4.7	Electric	Semi-Automatic	157495	2	2	7950
8	Kia	Sportage	2014	2.6	Hybrid	Manual	98700	3	4	9926
9	Toyota	RAV4	2005	3.1	Petrol	Manual	107724	2	5	6545

TASK 02 : Find the basic descriptive statistics using the `.describe()` method to get the mean, minimum, maximum, and standard deviation of the numeric columns only.

❖ Finding the basic descriptive statistics:

```
df.describe()
```

Output:

	Year	Engine_Size	Mileage	Doors	Owner_Count	Price
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	2011.543700	3.000560	149239.111800	3.497100	2.991100	8852.96440
std	6.897699	1.149324	86322.348957	1.110097	1.422682	3112.59681
min	2000.000000	1.000000	25.000000	2.000000	1.000000	2000.00000
25%	2006.000000	2.000000	74649.250000	3.000000	2.000000	6646.00000
50%	2012.000000	3.000000	149587.000000	3.000000	3.000000	8858.50000
75%	2017.000000	4.000000	223577.500000	4.000000	4.000000	11086.50000
max	2023.000000	5.000000	299947.000000	5.000000	5.000000	18301.00000

TASK 03 : Calculating the median of Mileage and Price using appropriate Pandas methods

```
df[['Mileage','Price']].median()
```

Output :

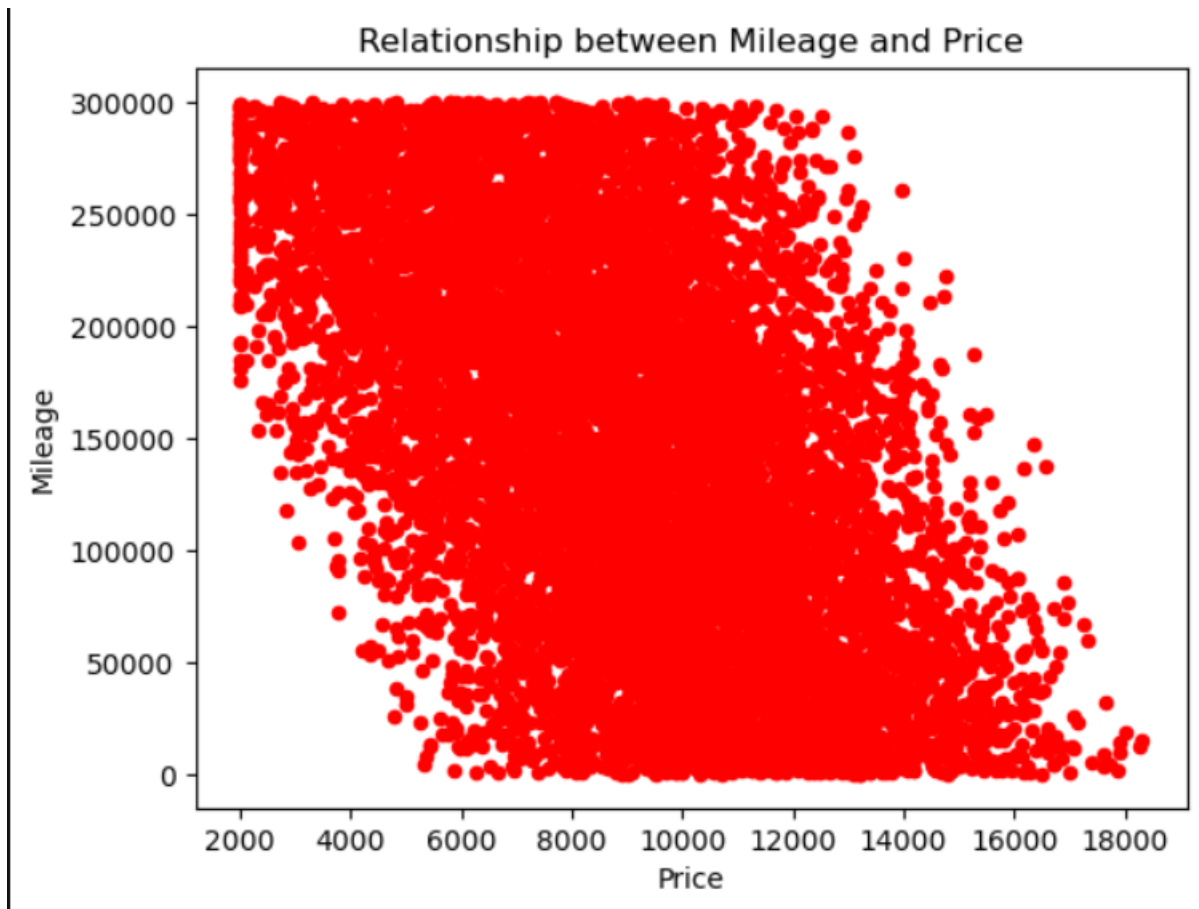
```
Mileage    149587.0
Price       8858.5
dtype: float64
```

TASK 04 : Analyzing Relationships between columns

4.1 – Creating a scatter plot with a meaningful title

```
df.plot.scatter(x = 'Price', y = 'Mileage', c = 'red', s = 15)
plt.title('Relationship between Mileage and Price')
```

Output:



4.2 – short paragraph about the plot :

- ❖ It is clear that when the mileage increases the price decreases. Which means the higher the mileage is, the lower the value of a vehicle goes. The conclusion is higher mileage cars are inclined to have low prices.

TASK 05 : Exploring Correlations

5.1 – Calculate the correlation between Milage and Price and between Engine_Size and Price

```
df['Mileage'].corr(df['Price'])
```

Output :

```
-0.5512271827629014
```

```
df['Engine_Size'].corr(df['Price'])
```

Output :

```
0.35740347535586653
```

5.2 – Explanation about what these correlations suggest about the relationship between price and these two attributes.

- The correlation between Mileage and Price is closer to -1. Therefore it indicates a negative relationship. Suggesting that when the price increases, the mileage decreases, and other way around.
- The correlation between Engine_Size and Price is closer to +1 than -1 therefore can be considered as a mildly positive relationship. Suggesting that when the price increases Engine_Size increases too.

PROBLEM 02

TASK 01 : Load the amazonFood.csv file into a Pandas DataFrame named "DF". Display the first five rows of the DataFrame and list all the columns. Provide a brief description of what information each column represents.

❖ Importing pandas and loading csv file.

```
import pandas as pd

DF = pd.read_csv('C:/Users/Shanya/Downloads/amazonFood.csv')
```

Output:

	ProductId	UserId	productName	Score	Summary	Text
0	B001E4KFG0	A3SGXH7AUHU8GW	Dogswell Vitality Canned Dog Food Case Duck	5	Good Quality Dog Food	I have bought several of the Vitality canned d...
1	B00813GRG4	A1D87F6ZCVE5NK	Southern Grove Premium Quality Roasted and Sal...	1	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...
2	B000LQOCH0	ABXLMWJIXXAIN	Turkish Delight Filbert (Hazelnuts) (Sultan) 1...	4	"Delight" says it all	This is a confection that has been around a fe...
3	B000UA0QIQ	A395BORC6FGVXV	Faeries Finest Flavor Extract, Cherry, 2.04 Ounce	2	Cough Medicine	If you are looking for the secret ingredient i...
4	B006K2ZZ7K	A1UQR5CLF8GW1T	Salt Water Taffy - Assorted, 5 lbs	5	Great taffy	Great taffy at a great price. There was a wid...
...	...	...	...	...	...	...
41705	B007IVE938	A3MDETK0D5UAIN	Ajika Organic Rose Buds, 1.6-Ounce	1	Not what I expected.	I certainly could do better by these if I just...
41706	B00125NESM	AGGS9IPJDU6ML	JOHN BOOS FBLG7224 Table,Stainless,Shelf,72"x24"	5	Great!!	Table is 35 inches in height. It is heavy dut...
41707	B003M5W8ZK	A373MY0PQL9R41	Green Mountain Coffee - Unsweetened Black Iced...	5	Black tea v Coca Cola	I have been one of those people that always fe...
41708	B003M5W8ZK	A3U2QUKGA7CJOP	Green Mountain Coffee - Unsweetened Black Iced...	1	Never Received the product	I was charged for this product by Liquid Natio...
41709	B005GXGAGY	A33AYVATA17JUR	LATE JULY CHIP MLTIGRN GRNMOJO ORG, 5.5 OZ, PK...	5	Great taste - and gluten free	Everyone in my family loves this chip - from t...

41710 rows x 6 columns

❖ Displaying first five rows

```
DF.head(5)
```

Output :

	ProductId	UserId	productName	Score	Summary	Text
0	B001E4KFG0	A3SGXH7AUHU8GW	Dogswell Vitality Canned Dog Food Case Duck	5	Good Quality Dog Food	I have bought several of the Vitality canned d...
1	B00813GRG4	A1D87F6ZCVE5NK	Southern Grove Premium Quality Roasted and Sal...	1	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...
2	B000LQOCH0	ABXLMWJIXXAIN	Turkish Delight Filbert (Hazelnuts) (Sultan) 1...	4	"Delight" says it all	This is a confection that has been around a fe...
3	B000UA0QIQ	A395BORC6FGVXV	Faeries Finest Flavor Extract, Cherry, 2.04 Ounce	2	Cough Medicine	If you are looking for the secret ingredient i...
4	B006K2ZZ7K	A1UQR5CLF8GW1T	Salt Water Taffy - Assorted, 5 lbs	5	Great taffy	Great taffy at a great price. There was a wid...

❖ Description of what each column represent:

- **ProductId** - A label to uniquely identify a certain product
- **UserId** - A label to uniquely identify the person who gave the feedback
- **productName** - Name of the product that has been given the feedback of
- **Summary** - Shows what the feedback is about
- **Text** - Complete feedbacks given by users

TASK 02 : Create a new column called `word_count` which stores the number of words in each review.

❖ Adding new column

```
DF['word_count'] = ''
```

❖ Calculating length

```
DF['word_count'] = DF['Text'].str.len()
```

DF

Output :

	ProductId	UserId	productName	Score	Summary	Text	word_count
0	B001E4KFG0	A3SGXH7AUHU8GW	Dogswell Vitality Canned Dog Food Case Duck	5	Good Quality Dog Food	I have bought several of the Vitality canned d...	263
1	B00813GRG4	A1D87F6ZCVE5NK	Southern Grove Premium Quality Roasted and Sal...	1	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...	190
2	B000LQOCH0	ABXLMWJIXXAIN	Turkish Delight Filbert (Hazelnuts) (Sultan) 1...	4	"Delight" says it all	This is a confection that has been around a fe...	509
3	B000UA0QIQ	A395BORC6FGVXV	Faeries Finest Flavor Extract, Cherry, 2.04 Ounce	2	Cough Medicine	If you are looking for the secret ingredient i...	219
4	B006K2ZZ7K	A1UQRSCLF8GW1T	Salt Water Taffy - Assorted, 5 lbs	5	Great taffy	Great taffy at a great price. There was a wid...	140
...	...	...	...	...	...	...	...
41705	B007IVE938	A3MDETK0D5UAIN	Ajika Organic Rose Buds, 1.6-Ounce	1	Not what I expected.	I certainly could do better by these if I just...	733
41706	B00125NESM	AGGS9IPJDU6ML	JOHN BOOS FBLG7224 Table,Stainless,Shelf,72"x24"	5	Great!!	Table is 35 inches in height. It is heavy dut...	484
41707	B003M5W8ZK	A373MY0PQL9R41	Green Mountain Coffee - Unsweetened Black Iced...	5	Black tea v Coca Cola	I have been one of those people that always fe...	285
41708	B003M5W8ZK	A3U2QUKGA7CJOP	Green Mountain Coffee - Unsweetened Black Iced...	1	Never Received the product	I was charged for this product by Liquid Natio...	186
41709	B005GXGAGY	A33AYVATA17JUR	LATE JULY CHIP MLTIGRN GRNMOJO ORG, 5.5 OZ, PK...	5	Great taste - and gluten free	Everyone in my family loves this chip - from t...	151

41710 rows × 7 columns

Task 03 : Creating a new column named `length_category` based on given conditions.

3.1 - Implementing it using a for loop, while loop and vectorization.

❖ For loop :

```
DF['length_category'] = ""
for index in DF.index :
    if DF.loc[index, 'word_count'] < 100 :
        DF.loc[index, 'length_category'] = 'short'
    elif DF.loc[index, 'word_count'] > 200 :
        DF.loc[index, 'length_category'] = 'long'
    elif 100 < DF.loc[index, 'word_count'] < 200 :
        DF.loc[index, 'length_category'] = 'medium'
```

❖ While loop :

```
DF['length_category'] = ""
i = 0
while i < len(DF):
    if DF.loc[i, 'word_count'] < 100 :
        DF.loc[i, 'length_category'] = 'short'
    elif DF.loc[i, 'word_count'] > 200 :
        DF.loc[i, 'length_category'] = 'long'
    elif (DF.loc[i, 'word_count'] >= 100) and (DF.loc[i, 'word_count'] <= 200) :
        DF.loc[i, 'length_category'] = 'medium'
    i = i + 1
DF
```

❖ Vectorisation

```
DF['length_category'] = ""
DF.loc[DF.word_count<100,'length_category'] = 'short'
DF.loc[DF.word_count>200,'length_category'] = 'long'
DF.loc[(DF.word_count>100) & (DF.word_count<200),'length_category'] = 'medium'
DF
```

Output for all three methods :

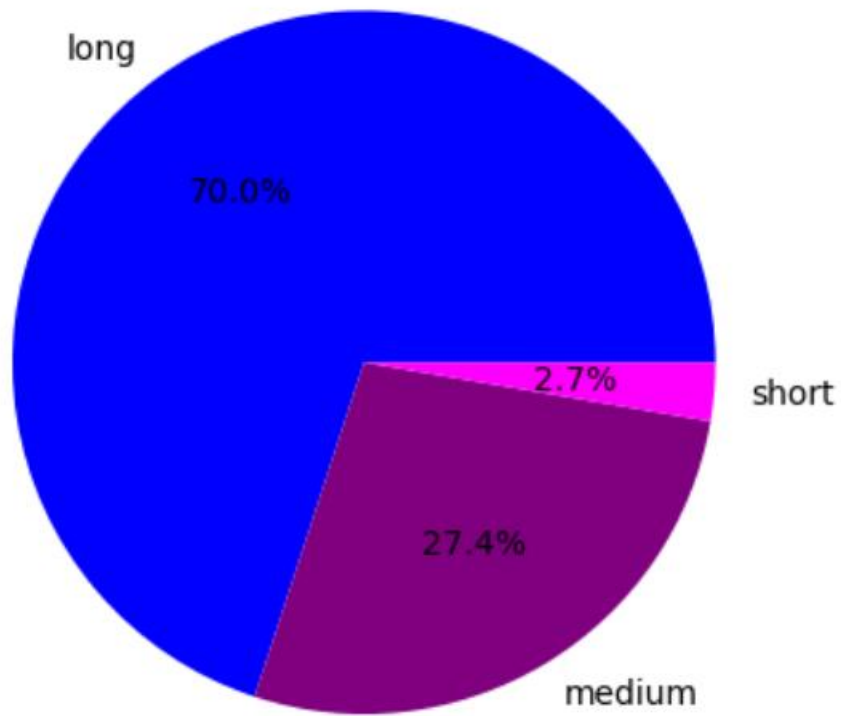
	ProductId	UserId	productName	Score	Summary	Text	word_count	length_category
0	B001E4KFG0	A3SGXH7AUHU8GW	Dogswell Vitality Canned Dog Food Case Duck	5	Good Quality Dog Food	I have bought several of the Vitality canned d...	263	long
1	B00813GRG4	A1D87F6ZCVE5NK	Southern Grove Premium Quality Roasted and Sal...	1	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...	190	medium
2	B000LQOCH0	ABXLMWJIXXAIN	Turkish Delight Filbert (Hazelnuts) (Sultan) 1...	4	"Delight" says it all	This is a confection that has been around a fe...	509	long
3	B000UA0QIQ	A395BORC6FGVXV	Faeries Finest Flavor Extract, Cherry, 2.04 Ounce	2	Cough Medicine	If you are looking for the secret ingredient i...	219	long
4	B006K2ZZ7K	A1UQRSLF8GW1T	Salt Water Taffy - Assorted, 5 lbs	5	Great taffy	Great taffy at a great price. There was a wid...	140	medium
...	...	...	...	...	...	...	...	...
41705	B007IVE938	A3MDET00D5UAIN	Ajika Organic Rose Buds, 1.6-Ounce	1	Not what I expected.	I certainly could do better by these if I just...	733	long
41706	B00125NESM	AGGS9IPJDU6ML	JOHN BOOS FBLG7224 Table,Stainless,Shelf,72"x24"	5	Great!!	Table is 35 inches in height. It is heavy dut...	484	long
41707	B003M5W8ZK	A373MY0PQL9R41	Green Mountain Coffee - Unsweetened Black Iced...	5	Black tea v Coca Cola	I have been one of those people that always fe...	285	long
41708	B003M5W8ZK	A3U2QUKGA7CJOP	Green Mountain Coffee - Unsweetened Black Iced...	1	Never Received the product	I was charged for this product by Liquid Natio...	186	medium
41709	B005GXGAGY	A33AYVATA17JUR	LATE JULY CHIP MLTIGRN GRNMOJO ORG, 5.5 OZ, PK...	5	Great taste - and gluten free	Everyone in my family loves this chip - from t...	151	medium

41710 rows × 8 columns

3.2 – Plotting a pie chart showing the proportion of each length_category.

```
import matplotlib.pyplot as plt
variable = DF.length_category.value_counts()
plt.pie(variable, labels = ['long', 'medium', 'short'], colors=['blue', 'purple', 'magenta'], autopct='%1.1f%%')
```


Output :



PROBLEM 03

TASK 01 : Load the "college_major" dataset using the URL provided and display its structure and load first few records

❖ Loading csv file

```
import pandas as pd

df1 = pd.read_csv('https://raw.githubusercontent.com/fivethirtyeight/data/master/college-
majors/recent-grads.csv')

df1
```

Output:

5] :

	Rank	Major_code	Major	Total	Men	Women	Major_category	ShareWomen	Sample_size	Employed	...	Part_time	Full_time_year_round	Unempl
	0	1	2419	PETROLEUM ENGINEERING	2339.0	2057.0	282.0	Engineering	0.120564	36	1976	...	270	1207
	1	2	2416	MINING AND MINERAL ENGINEERING	756.0	679.0	77.0	Engineering	0.101852	7	640	...	170	388
	2	3	2415	METALLURGICAL ENGINEERING	856.0	725.0	131.0	Engineering	0.153037	3	648	...	133	340
	3	4	2417	NAVAL ARCHITECTURE AND MARINE ENGINEERING	1258.0	1123.0	135.0	Engineering	0.107313	16	758	...	150	692
	4	5	2405	CHEMICAL ENGINEERING	32260.0	21239.0	11021.0	Engineering	0.341631	289	25694	...	5180	16697
	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	168	169	3609	ZOOLOGY	8409.0	3050.0	5359.0	Biology & Life Science	0.637293	47	6259	...	2190	3602
	169	170	5201	EDUCATIONAL PSYCHOLOGY	2854.0	522.0	2332.0	Psychology & Social Work	0.817099	7	2125	...	572	1211
	170	171	5202	CLINICAL PSYCHOLOGY	2838.0	568.0	2270.0	Psychology & Social Work	0.799859	13	2101	...	648	1293
	171	172	5203	COUNSELING PSYCHOLOGY	4626.0	931.0	3695.0	Psychology & Social Work	0.798746	21	3777	...	965	2738
	172	173	3501	LIBRARY SCIENCE	1098.0	134.0	964.0	Education	0.877960	2	742	...	237	410

173 rows × 21 columns

❖ Displaying first few rows and columns

```
df1.head()
```

Output :

	Rank	Major_code	Major	Total	Men	Women	Major_category	ShareWomen	Sample_size	Employed	...	Part_time	Full_time_year_round	Unemploy
0	1	2419	PETROLEUM ENGINEERING	2339.0	2057.0	282.0	Engineering	0.120564	36	1976	...	270	1207	
1	2	2416	MINING AND MINERAL ENGINEERING	756.0	679.0	77.0	Engineering	0.101852	7	640	...	170	388	
2	3	2415	METALLURGICAL ENGINEERING	856.0	725.0	131.0	Engineering	0.153037	3	648	...	133	340	
3	4	2417	NAVAL ARCHITECTURE AND MARINE ENGINEERING	1258.0	1123.0	135.0	Engineering	0.107313	16	758	...	150	692	
4	5	2405	CHEMICAL ENGINEERING	32260.0	21239.0	11021.0	Engineering	0.341631	289	25694	...	5180	16697	16

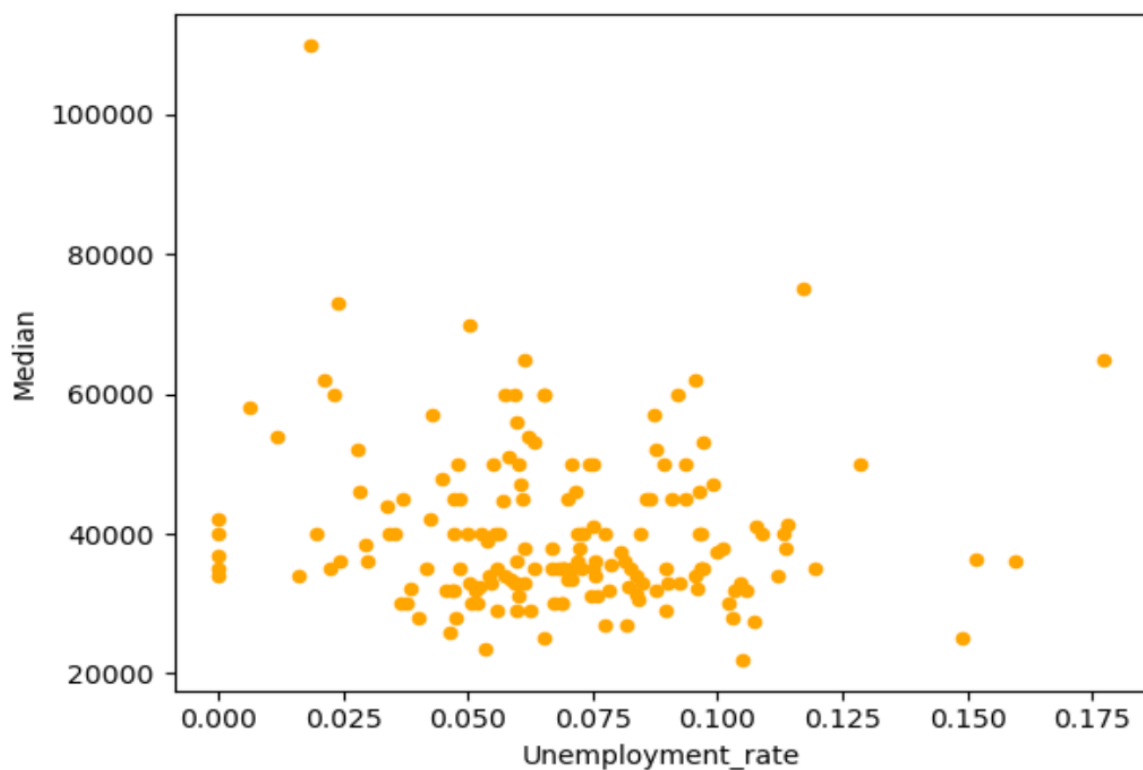
5 rows × 21 columns

TASK 02 : Identifying whether there is a general trend between earnings and unemployment using a scatter plot

```
import matplotlib.pyplot as plt
df1.plot('Unemployment_rate', 'Median', kind = 'scatter', color='orange')
df1['Median'].corr(df1['Unemployment_rate'])
```

Output :

-0.1088332938973364



Analysis –

- The correlation is negative as the value is closer to -1. Suggesting that when the median salary increases the unemployment rate decreases,
- Indicating that if a major with higher median salary is chosen, lower the chance of unemployment will be.

TASK 03 : Determining which specific majors offer both high salaries and low unemployment rates.

```
df1[(df1["Median"] > 60000) & (df1["Unemployment_rate"] < 0.05)]
```

Output :

Rank	Major_code	Major	Total	Men	Women	Major_category	ShareWomen	Sample_size	Employed	...	Part_time	Full_time_year_round	Unemployed	
0	1	2419	PETROLEUM ENGINEERING	2339.0	2057.0	282.0	Engineering	0.120564	36	1976	...	270	1207	37
2	3	2415	METALLURGICAL ENGINEERING	856.0	725.0	131.0	Engineering	0.153037	3	648	...	133	340	16
7	8	5001	ASTRONOMY AND ASTROPHYSICS	1792.0	832.0	960.0	Physical Sciences	0.535714	10	1526	...	553	827	33

3 rows × 21 columns

Analysis –

- According to the output , PETROLEUM ENGINEERING, METALLURGICAL ENGINEERING AND ASTRONOMY AND ASTROPHYSICS offer both high salaries and low unemployment rates.

TASK 04 : Grouping the data by Major_category and calculate the average Median salary for each category and display the results.

```
grp_data = df1.groupby("Major_category")  
data_avg = grp_data["Median"].mean()  
data_avg
```

Output :

Major_category	
Agriculture & Natural Resources	36900.000000
Arts	33062.500000
Biology & Life Science	36421.428571
Business	43538.461538
Communications & Journalism	34500.000000
Computers & Mathematics	42745.454545
Education	32350.000000
Engineering	57382.758621
Health	36825.000000
Humanities & Liberal Arts	31913.333333
Industrial Arts & Consumer Services	36342.857143
Interdisciplinary	35000.000000
Law & Public Policy	42200.000000
Physical Sciences	41890.000000
Psychology & Social Work	30100.000000
Social Science	37344.444444

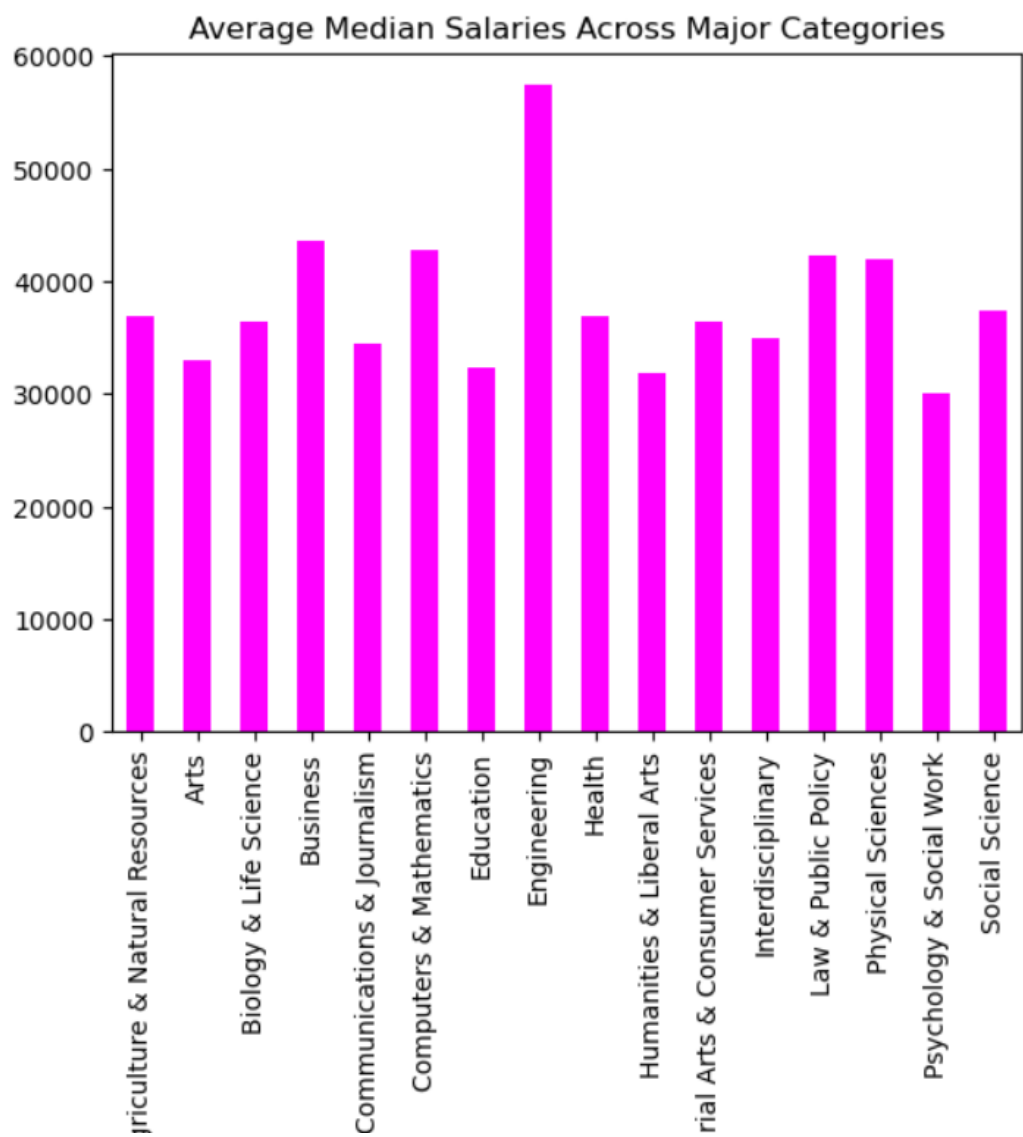
Name: Median, dtype: float64

TASK 05 :

Creating a bar chart to visualize the average median salaries across different Major_category groups providing meaningful labelling and title to the chart.

```
data_avg.plot(kind = 'bar', color = 'magenta')  
plt.title('Average Median Salaries Across Major Categories')
```

Output :



PROBLEM 04

TASK 01 :

Writing a Python Program to create an SQLite database named Hospital_System.db, and create the listed Hospital and Ward table.

❖ Importing pandas , sqlite3 and creating cursor to execute queries.

```
import pandas as pd
import sqlite3 as sq3
conn = sq3.connect('Hospital_System.db')
cur = conn.cursor()
```

❖ Creating hospital and ward table

```
create_Hospital = """CREATE TABLE Hospital (
    hospital_id INT(3),
    address VARCHAR(50) UNIQUE NOT NULL,
    zip_code VARCHAR(10) NOT NULL,
    city VARCHAR(50) NOT NULL,
    region VARCHAR(50) NOT NULL,
    country VARCHAR(50) NOT NULL,
    CONSTRAINT hospital_pk PRIMARY KEY (hospital_id)
);"""

create_Ward = """CREATE TABLE Wards (
    ward_id INT(4),
    ward_name VARCHAR(20) UNIQUE NOT NULL,
    hospital_id INT(3),
    CONSTRAINT ward_pk PRIMARY KEY (ward_id),
    CONSTRAINT hospital_fk FOREIGN KEY (hospital_id)
    REFERENCES branches(ward_id)
);"""
```

❖ Creating hospital table

```
cur.execute(create_Hospital)
```

❖ Creating ward table

```
cur.execute(create_Ward)
```

- ❖ **TASK 02** : Inserting the provided data into the Hospital and Ward tables.

```
hospital_info = [(101, '50 Green Street', 'SW1 2AA', 'London', 'London', 'UK'),  
                 (102, '12 Hill Avenue', 'M1 5GH', 'Manchester', 'Greater Manchester', 'UK'),  
                 (103, '34 River Road', 'L2 3DR', 'Liverpool', 'Merseyside', 'UK'),  
                 (104, '19 Lake View', 'B4 2DD', 'Birmingham', 'West Midlands', 'UK')]
```

```
ward_info = [(1010, 'Accident', 101),  
             (1020, 'Cardiology', 102),  
             (1030, 'Endoscopy', 103),  
             (1040, 'Surgical', 104),  
             (1050, 'Emergency', "")]
```

```
cur.executemany('Insert into Hospital values(?,?,?,?,?)', hospital_info)
```

```
cur.executemany('Insert into Wards values(?,?,?)', ward_info)
```


TASK 03 : Display all wards along with their Hospital locations, and all Hospital without any Wards.

❖ **All wards with their hospital locations.**

```
cur.execute("""SELECT w.ward_name, h.address, h.city, h.region, h.country
              FROM Wards w
              LEFT JOIN Hospital h
              ON w.hospital_id = h.hospital_id""")

cur.fetchall()
```

Output :

```
[('Accident', '50 Green Street', 'London', 'London', 'UK'),
 ('Cardiology', '12 Hill Avenue', 'Manchester', 'Greater Manchester', 'UK'),
 ('Endoscopy', '34 River Road', 'Liverpool', 'Merseyside', 'UK'),
 ('Surgical', '19 Lake View', 'Birmingham', 'West Midlands', 'UK'),
 ('Emergency', None, None, None, None)]
```

❖ **All hospitals without any wards**

```
cur.execute("""SELECT h.hospital_id
              FROM Hospital h
              LEFT JOIN Wards w
              ON h.hospital_id = w.hospital_id
              WHERE w.ward_id IS NULL""")

cur.fetchall()
```

Output :

```
[]
```

➤ **There are no hospitals without any wards.**

